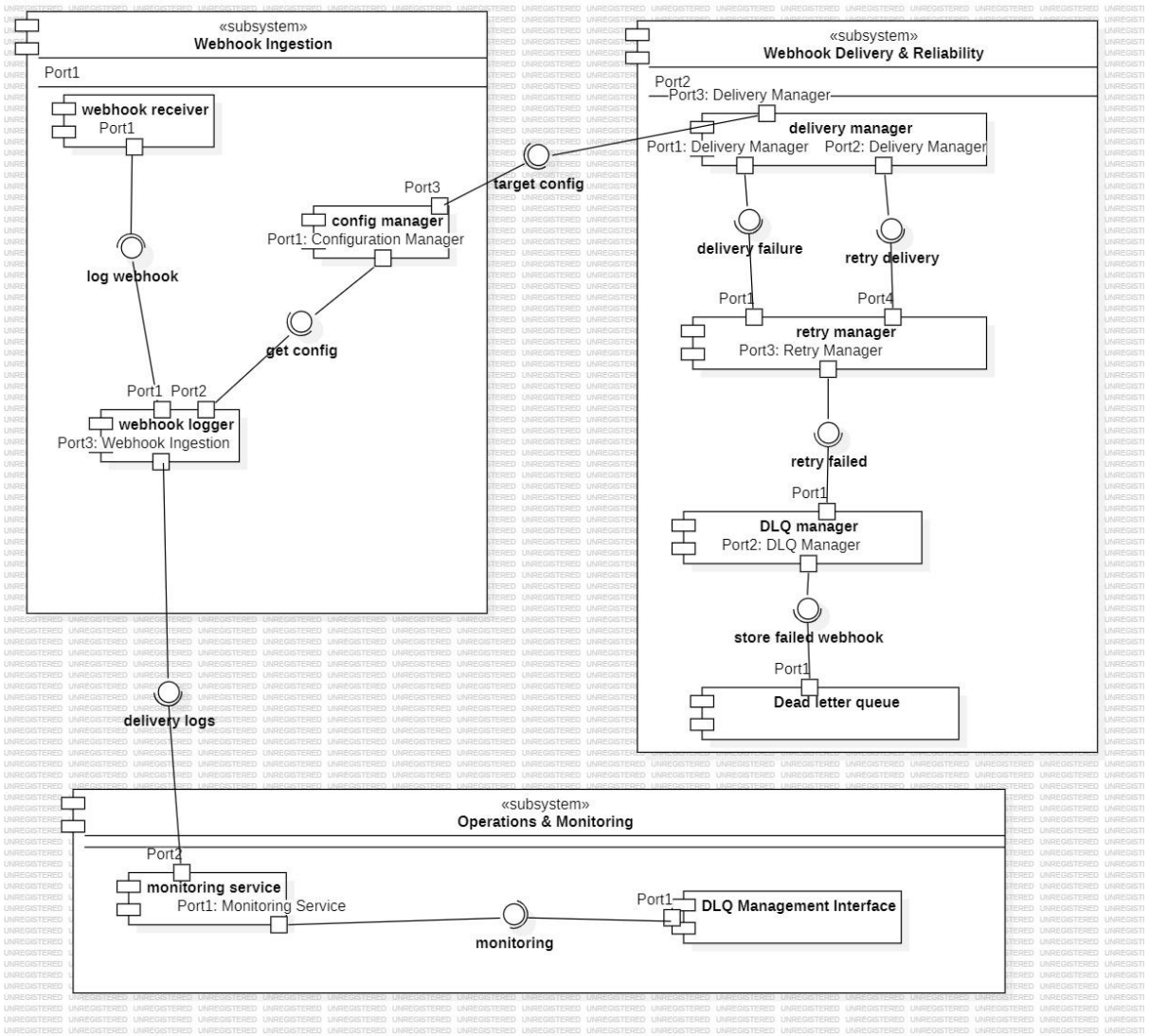
**Name: Shreyas B Patil SRN: PES2UG24CS488 SEC:H LAB:3**

**Component diagram:**




**JUSTIFICATION:**

**Architecture Selection: Microservices Architecture**

**Architectural Choice:**
The system is divided into independent services/components such as **Webhook Ingestion, Webhook Delivery & Reliability, and Operations & Monitoring**, with each service responsible for a specific function.

**Two main reasons:**

1. **Independent failure handling:** Webhook delivery failures and retry processing can be handled separately from webhook ingestion, so a temporary failure in delivery does not stop the system from receiving new webhooks.
2. **Independent scalability:** Components such as the Webhook Receiver and Delivery Manager can be scaled independently when webhook traffic or delivery workload increases.

**Security Advantage:**
Each service can have separate access controls and permissions, limiting what a compromised service can access. For example, the DLQ-related service can be restricted to dead-letter data and management operations.

**Performance Benefit:**
Independent services can be scaled horizontally according to workload. For example, multiple Webhook Receiver instances can process incoming webhook requests concurrently during high traffic, improving throughput.